Capstone Project Write-Up

Emergency Shelter & Housing Outcomes - Factors for success

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Table of contents

1 Title, Team, and Affiliations

Project title: Emergency Shelter & Housing Outcomes - Factors for success

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Repository: `https://github.com/stuckerWU/data510-capstone`

2 Abstract

Problem: Church at the Park (C@P) provides vital services to the local community, and generates data from multiple locations to provide those services, but does not have their data sources integrated. This lack of integration makes it difficult for C@P to analyze their data to gain insight into what factors drive positive outcomes. Through this work we hoped to help C@P's staff better understand their program outcomes and advocate for increased resources that will lead to more community benefit.

We integrated three different C@P data sources, and investigated factors associated with the wait time to enter shelter, and how long it takes to exit shelter. To accomplish this, we used statistical analysis methods that examine duration of time until certain events occur, and built a machine learning model aimed at predicting outcomes of applications and exits to shelter.

With our duration analysis and machine learning models, we identified a variety of variables that are associated with the time to enter or exit shelter. These variables tended to fall into two groups: specific circumstances or traits of a participant, and engagement with case management.

Based on our exploration of the data and our discussions with C@P, we also found that real world circumstances that are not captured in C@P's data may also affect the results and interpretation. Of particular note, we found that the volume of applications for shelter significantly outpaces the availability of shelter. Only 14% of applications could ultimately be matched to a stay with C@P, showing that there is a significant need for shelter in the Salem area community. Additionally, we identified a variety of opportunities where C@P may want to

invest resources to improve outcomes or adjust processes to improve data collection for future analysis.

3 Introduction: Problem and Research Questions

3.1 Problem

Church at the Park (C@P) provides vital services to the local community, including food distribution, educational programs, and social support. However, the organization faces challenges in efficiently managing and analyzing the data related to these services, which limits their ability to make data-driven decisions, improve service delivery, and make a strong case for additional monetary support.

3.2 Research Questions

1. Can we construct a sustainable data engineering pipeline that integrates C@P's 3 program datasets for comprehensive analysis and reporting?
2. Can we derive insights from integrated data to improve service delivery and support decision-making? Specifically:

   2a. Are there factors that are correlated to the average length of wait to get into shelter, and the average stay in shelter?

   2b. Are there factors that are correlated to whether an individual is more likely to return to C@P for services?

   2c. Are there factors correlated with successful applications for assistance?

4 Impact and Stakeholders

4.1 Who Benefits

The primary beneficiaries of this project are the staff and leadership of C@P, who will gain a more comprehensive understanding of their service delivery and client outcomes through the integrated data pipeline. This will enable them to make informed decisions about resource allocation, program improvements, and strategic planning. Additionally, the individuals who rely on C@P's services may indirectly benefit from improved service delivery and outcomes resulting from data-driven insights.

4.2 Organizational or Societal Impact

By improving C@P's ability to analyze and utilize their data, this project has the potential to enhance the effectiveness of their programs, leading to better support for homeless individuals in Salem. This could contribute to broader societal benefits by helping to address homelessness more effectively and efficiently.

5 Related Work and Background

5.1 Background on C@P

- C@P is a nonprofit organization based in Salem, Oregon that provides services for unsheltered neighbors. Their services include providing vital low barrier shelter, and employment services. This project focuses on C@P's five emergency shelter programs. C@P describes their sheltering programs as follows:

  "Low-barrier sheltering is provided through non-congregate communities where we offer emergency housing for people of all ages, regardless of race, religion, gender, gender expression, age, national origin, disability, sexual orientation, sobriety, or mental health. We allow people to remain with pets, partners, and property, which helps to remove a barrier that would often preclude them from seeking shelter or services elsewhere. C@P works alongside guests to exit shelter into more stable housing." (From https://www.church-at-the-park.org/about)

- C@P's five different emergency shelter programs serve different populations and provide different services.

Location	Notes	Capacity
Village of Hope (VOH)	A shelter for participants ages 18 and above, opened before 2024.	Has 80 beds available.
Family Shelter (CCS)	A shelter is for households with children under the age of 18. opened before 2024	Has 44 pods for families experiencing homelessness (135 beds available)
Young Adult Shelter (YAS)	A shelter for participants ages 18 - 24, opened before 2024.	Has 21 beds available.
Polk County Site (also referred to as Monmouth, or MON)	A shelter for participants ages 18 and above. Has some requirements for participants to be eligible (for example, ties to rural Polk County are required.) Opened June 2025.	Has 20 beds available.

Location	Notes	Capacity
Safe Parking (SP)	Gives participants living in their car a safe place of park each night. Opened before 2024.	11 spots

- C@P utilizes federal and state mandated databases for tracking stays, demographics, assessments, and case notes. Many of the fields are defined by the U.S. Department of Housing and Urban Development, and we relied on the organization's definitions for program exit types and their program guidelines.

5.2 Similar Work

There are many studies that analyze homelessness, including the factors that lead to either entering or exiting homelessness. We identified a few where the authors conducted research similar to our own or relate to trends we found in our results.

- A 2019 study from University of Utah examined factors behind why families exit and return to shelters in Salt Lake County. This study used a similar statistical approach to our study and also focused on working with a single shelter. Of note, they found that subsidized housing program enrollment was connected to a reduction in the likelihood of an individual leaving and returning to shelter. Additionally, a family's prior income, prior residence, and exit from shelter destination also played a part in whether they return to shelter, and how long they stay. (Kim & Garcia, 2019)
- A 2020 study from Employment and Social Development Canada utilized data from 174 emergency shelters to examine pathways and housing outcomes. This study compared individuals who were using an emergency shelter for the first time versus those who were returning. Notably, they found that there does appear to be a difference in outcomes between those two groups. For example, certain entry reasons were associated with positive effect on the likelihood of leaving for acquired housing, such as entering as part of family. Additionally, they found that first time shelter users were five times more likely to exit to acquired housing compared to those who were returning. (Chen et.al, 2022)
- A 1997 study of New York City shelter usage examined how family characteristics were predictors of shelter exit and reentry. Those factors included family size, age, race and ethnicity, and public assistance receipt. The study also identified how city policy regarding eligibility for subsidized housing may have influenced how long individuals stayed in shelter. (Wong et al., 1997)
- In 2023, United Kingdom researchers conducted a meta-analysis on studies of case management programs for people experiencing homelessness. Broadly, the study found a connection between providing any type of case management, and better housing outcomes.

More specifically, intensive case management may be linked to more time in stable housing. (Weightman et. al, 2023)

Our literature review only briefly touches on the depth of research on homelessness. However, it's clear a number of factors can influence a person's outcomes, including personal traits, public policy, and services offered.

5.3 Technical Background

Latent Dirichlet Allocation (LDA) is a generative statistical model that allows sets of observations to be explained by unobserved groups that explain why some parts of the data are similar. LDA is often used in natural language processing to identify topics in a set of documents. In our case, we used LDA to identify topics in case notes, which are free text fields that case managers fill out for each participant. https://www.jmlr.org/papers/volume3/blei03a/blei03a.pdf

5.4 Data Dictionaries

C@P provided a data dictionary for the federal reporting data, which is also available online from the U.S. Department of Housing and Urban Development https://www.hudexchange.info/programs/hmis/hmis-data-standards/standards/introduction-to-hmis-data-standards/. For the other two data sources, we created a data dictionary based on our exploration of the data, and our conversations with C@P staff. The data dictionaries are included in the repository.

6 Data and Data Engineering

6.1 Data Sources

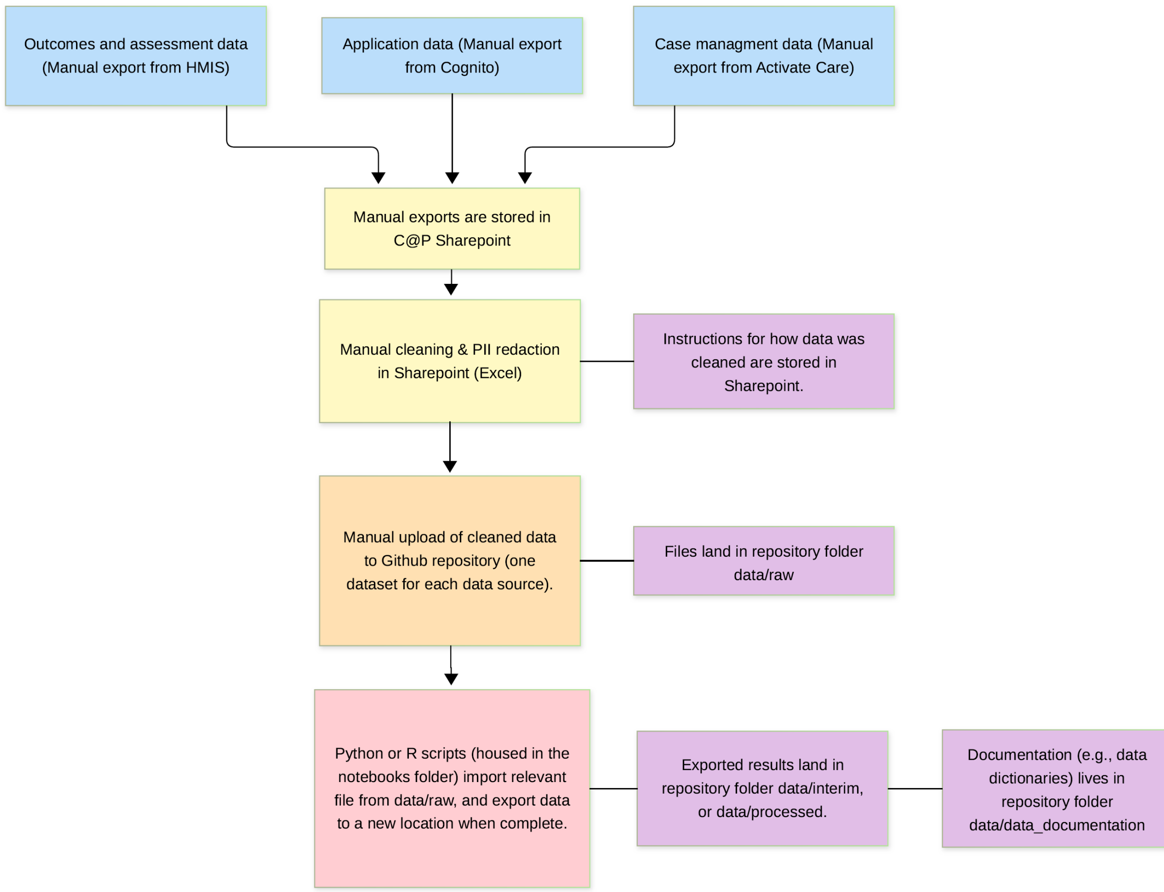
Our data came from three sources, all provided by C@P, for activity 2024-2025. A summary of our data sources is included below.

Source	Volume	Coverage	Access	Initial repository location
Application data (From Cognito)	6604 rows	2024-2025	Provided by C@P, a one-time manual pull completed by C@P	data/raw

Source	Volume	Coverage	Access	Initial repository location
Case management data (From Activate Care)	20,930 rows	2024-2025	Provided by C@P, a one-time manual pull completed by C@P	data/raw
Federal reporting data (From Homeless Information Management System, HMIS)	1,347 rows	2024-2025	Provided by C@P, a one-time manual pull completed by C@P	data/raw

6.2 Engineering Summary

Due to the confidentiality of the data and technical limitations in setting up access to it, our data pipeline required manual extraction and transformation. Our pipeline consisted of three major steps: (1) Manually extract data from C@P's three systems (C@P staff assisted with obtaining these extracts), (2) Manually cleaning, wrangling, and redacting confidential information in C@P's SharePoint, and (3) Exporting cleaned data to our GitHub repository. We then used Python and R scripts to analyze the data in GitHub and produce our deliverables.



One major limitation we discovered during step 2 of this process was the limited availability of matchable data points between the Applications data and the HMIS data. As such, we were only able to match a fraction of the total HMIS stays to their corresponding application. Additionally, this manually process was exacerbated by not having more powerful analysis tools to automatically find and evaluate potential duplicates en masse. This work could be greatly sped up by introducing some back-filling of the HMIS Record IDs to the applications data once an applicant has been approved for and moved into emergency shelter.

7 Ethics and Responsible Use

7.1 Privacy

We used data provided by C@P, with their consent. Prior to accessing data, team members filled out applications to serve as volunteers and consented to C@P performing background checks.

However, the manual data exports C@P provided contain personally identifiable information (PII). In our early conversations with C@P, we established and agreed to redact any PII before moving data into our GitHub repository. This ensures that sensitive information about the participants C@P assists are anonymized. Our process to anonymize the data took place in a few stages.

1) The data provided by C@P is initially stored in a SharePoint access point that C@P owns and controls access to. Thus, only C@P and our three team members have access to sensitive data.
2) We redacted the data that clearly identifies an individual (e.g., name, email, phone).
3) We masked columns that were not strictly PII, but contained sensitive information that could make the person more identifiable. We did this in two ways.
    1) By generalizing specific answers to a flag that indicates whether the person had an answer.

       Example 1: If the original response was "I stayed at Main Street last night", the generalized answer is simply 1, to indicate the individual provided a response.

       Example 2: If the original answer to age was "26", the generalized column puts the person into an age group, such as "25-29."

       Example 3: For a case note with narrative information about the individual's medical appointments or services rendered, we do not ultimately keep the case note, but instead just flag whether terms such as "medical" or "appointment" were in the case note to indicate types of assistance provided. All other details are stripped.
    2) By changing the name of certain columns with sensitive information (i.e., the answers themselves are not identifiable, but in combination with other information, such as age or gender, it could be easier to identify the individual). For example, with a column that answers whether the individual has a specific medical condition (thus, answers are just yes or no), we changed the name of the column. The true name of the column is only contained in Sharepoint and will only be shared with CAP in any analysis. For external facing analysis, we only used the masked name.

7.2 Fairness / Equity

Where reasonable, for participants with transgender identities, we folded them into broader male and female identities to preserve privacy. If viewed in combination with the race/ethnicity fields, age, and given the small number of participants with those identities, we noted it could become easy to identify specific individuals in C@P programs. This means our analysis is not granular enough to address whether there are differences in time frames or outcomes for all gender identities. However, we felt it was more important to preserve the privacy of these participants.

7.3 Limitations of Setting

Our analysis only focuses on one organization and one region in Oregon. In our discussion section, we address how this affects interpretation of our results.

7.4 Non-causal Framing

Our analysis does not focus on causation, and our discussion section addresses how other factors outside the data may influence interpretation.

8 Methods and Evaluation

8.1 Design

To explore the data, we calculated common descriptive statistics and plotted our data in a variety of ways (e.g., distributions, bar plots) to conduct exploratory data analysis.

For more in-depth analysis, we chose to implement two methods: survival analysis and machine learning.

8.1.1 Survival Analysis

Survival analysis is a statistical method that specifically examines how long it takes for an event to occur and helps scientists develop estimates for when an event will occur. Often, survival analysis is used in medical or industrial research. For our purposes, we are using this method to examine how long participants take to enter and exit shelter, and explore which variables modify the time it takes to enter or exit. In this write up, going forward, we will refer to survival analysis as a duration analysis, as we felt duration is more descriptive of the analysis we were doing, and more appropriate considering the focus of our project.

8.1.2 Machine Learning (XGBoost)

Machine learning algorithms help uncover patterns in data and can be used to develop models to predict specific results (for example, whether a customer will churn, or how long they will take to churn.) We opted to use machine learning to predict the status of a participant's application (successful, returned, or unsuccessful), and what kind of outcome a participant exited to, once they left shelter. We specifically chose to use the XGBoost library, which uses multiple decision trees as the basis of its machine learning model. We used this method because it supports missing values without the need for imputation. We often found missingness throughout our data, so XGBoost was a good fit, especially given that the missingness may simply be because the participant was not required to answer every question in full.

8.2 Validation/Evaluation Methods

8.2.1 Duration Analysis

For our duration analysis, our validation and evaluation involved a few steps.

- Generating Kaplan-Meier plots, which are a descriptive version of duration analysis, to review the shape, distribution, and general trends for time to enter or exit shelter.
- Developing duration regression models (based on the Weibull distribution) to model a smooth estimate of time to enter or exit. We developed models for nearly all of our variables, to review them individually.
- We examined p values from the regression model, to see if different answers for a given variable had significantly different duration curves.
- We overlaid our parametric Weibull duration curve models on top of our Kaplan-Meier plots to examine how consistent the model was with the raw data. Additionally, we reviewed the confidence intervals for Kaplan-Meier and noted how well our model fit within those intervals (or not).
- We calculated the Event to Time Ratio (ETR) for each set of covariates to quantitatively describe the difference in the Weibull curves.

8.2.2 Entry Into Shelter - Machine Learning (XGBoost)

The shelter entry model used a 5-fold cross validation approach, with stratification by the outcome variable and grouping by participant ID to ensure that all data for a given participant was contained within a single fold.

The shelter entry model used 80% of data for training each tree, and 20% of data for validation. The model was evaluated using a 5-fold cross validation approach, with stratification by the outcome variable and grouping by participant ID to ensure that all data for a given participant was contained within a single fold.

8.2.3 Exit Type - Machine Learning (XGBoost)

The final model dropped `other` as an outcome, and focused on a binary classification problem. The model was evaluated using a 5-fold cross validation approach, with stratification by the outcome variable and grouping by participant ID to ensure that all data for a given participant was contained within a single fold. The model was evaluated using a variety of metrics, including AUC, accuracy, balanced accuracy, and class-level precision and recall.

The exit model used 80% of data for training each tree, and 20% of data for validation. The model was evaluated using a 5-fold cross validation approach, with stratification by the outcome variable and grouping by participant ID to ensure that all data for a given participant was contained within a single fold.

9 Results and Visualizations

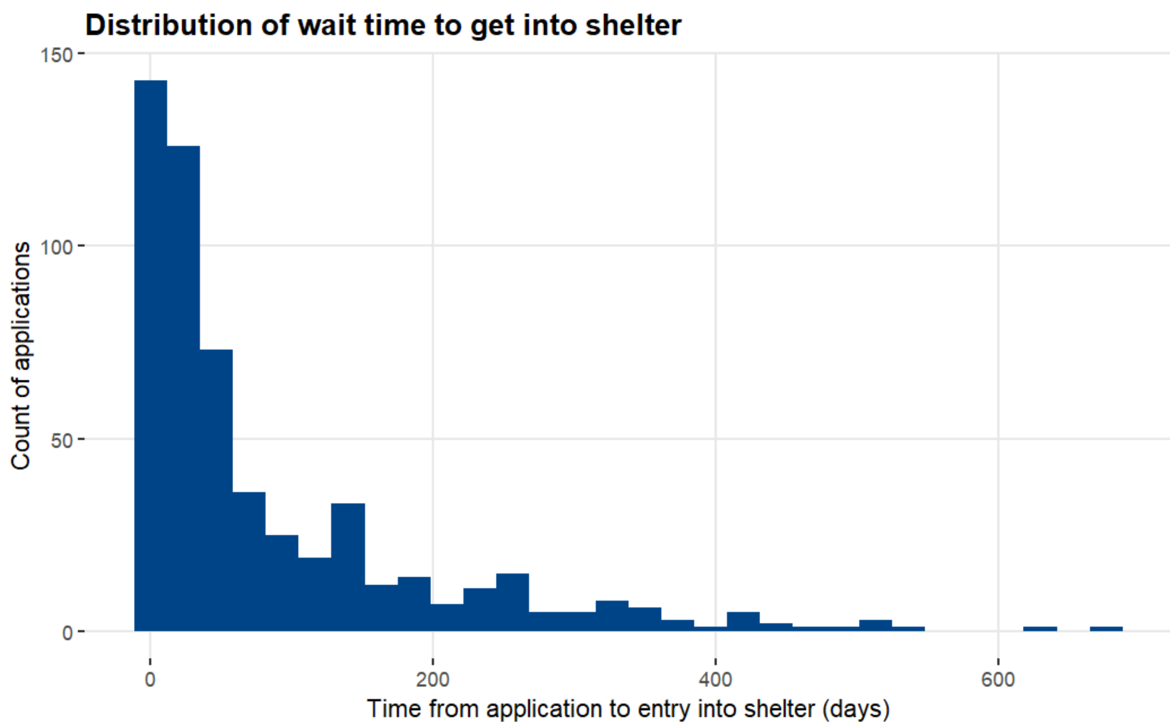
9.1 Notable Exploratory Analysis

9.1.1 Applications - Summary Information

Program	Number of applications	Percent of all applications
All	5392	
CCS	849	15.75%
YAS	560	10.39%
SP	920	17.06%
VOH	2895	53.69%
MON	168	3.12%

Out of 5,392 applications, we were able to match 768 applications to an id in the HUD tracking system (HMIS). Of those applications, 442 were the first successful applications, and 209 were return applications. The following table provides summary statistics for first successful applications.

Statistic	Time from application to entry into shelter
Mean	61 days
Median	27 days
Maximum	637 days
Minimum	Zero days



9.1.2 Stays - Summary Information

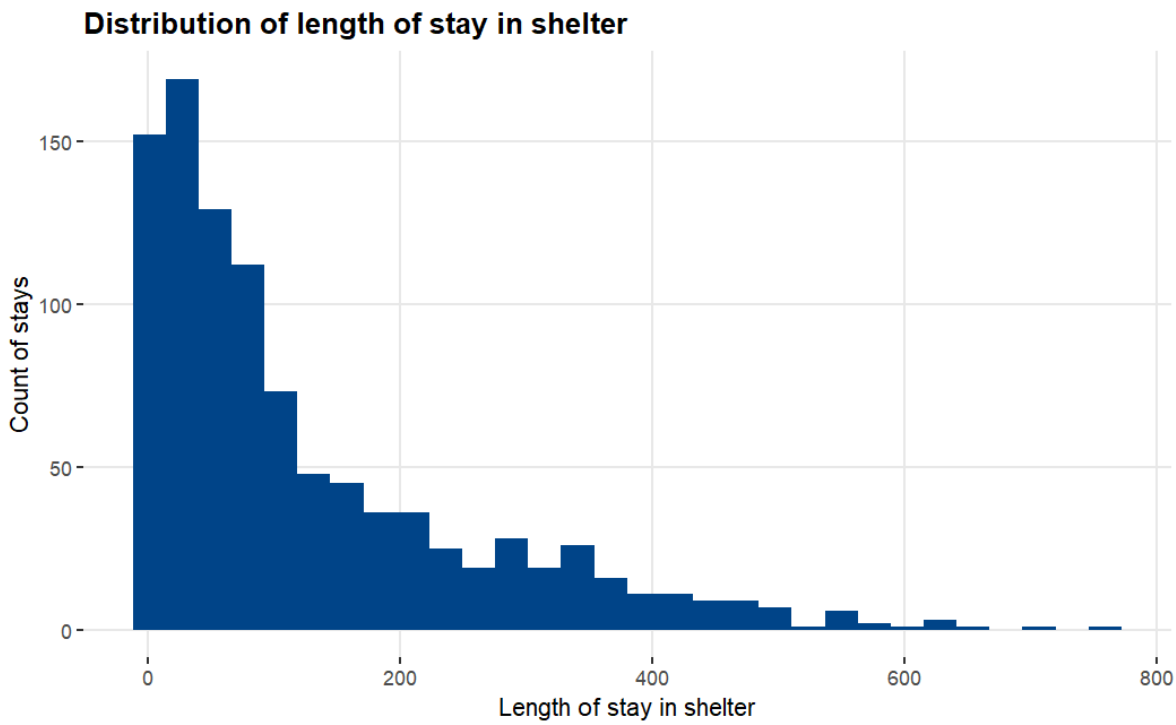
Program	Number of stays	Percent of all stays (based on stays with exit date)
All	996 (with exit date, and stay greater than zero days) 1108 (includes participants who have not exited yet)	
CCS	423	42.47%
YAS	133	13.35%
SP	209	20.98%
VOH	219	21.99%
MON	12	1.20%

Exit outcome	Mean Length of stay	Median Length of stay	Maximum Length of stay	Minimum Length of stay	Number of stays
All	127	77	759	1	996**

Exit outcome	Mean Length of stay	Median Length of stay	Maximum Length of stay	Minimum Length of stay	Number of stays
Positive	141	93	705	1	567*
Negative	110	55	759	1	394*
Other	72	17	439	3	34*

*Excludes rows where the exit date is NA (i.e., the participant hasn't exited yet) or length of stay is zero.

**Includes a stay where there was an exit date, but no exit destination.



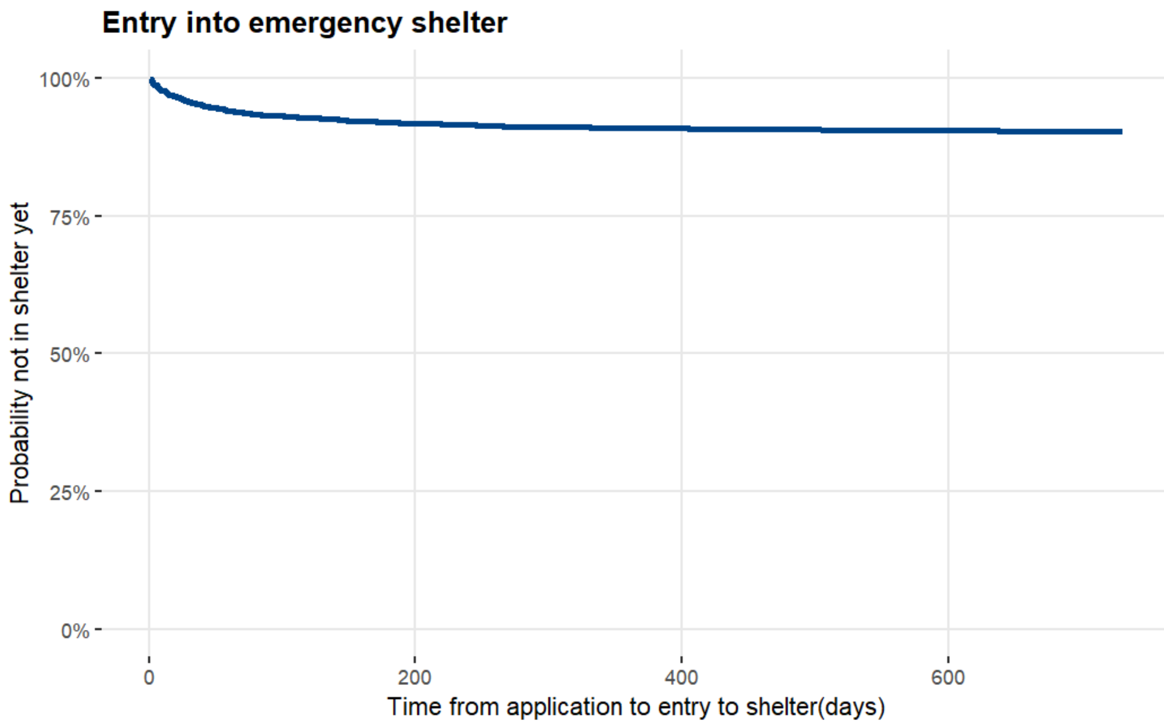
9.2 Duration Analysis

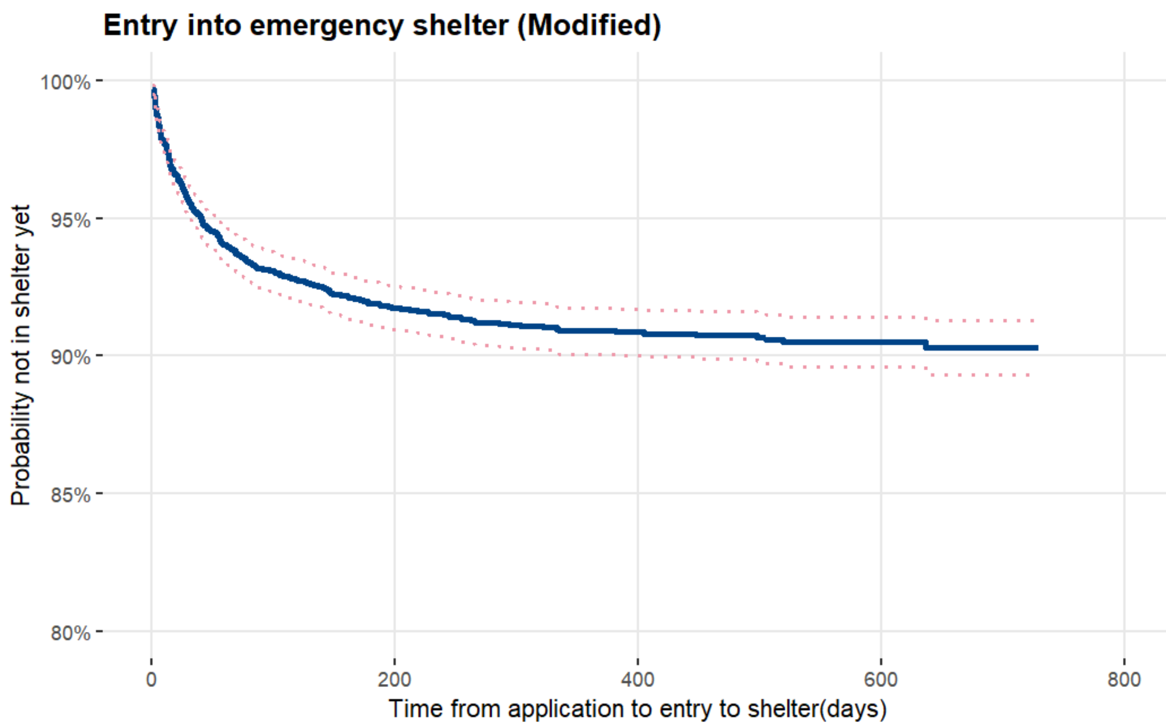
9.2.1 Housing Applications

We applied a duration analysis method to all applications, and to stratifications of specific variables. In general, we found multiple factors may affect the time to enter shelter. However, it's important to consider those findings in context. Specifically, we found it notable that the number of people who enter shelter after applying is very small. We were only able to match

768 applications to an id in the HMIS tracking system, which is only about 14% of applications. Of those applications, only 442 were designated as the first successful application (rather than a return application, or an earlier, unsuccessful application).

As a result, when we initially plotted a Kaplan-Meier duration curve, we see little change in the probability that a participant has entered shelter over time. The following figures show the curve using a 0 to 100% y axis, and an alternative zoomed in version that shows the curve from 80 to 100% on the y axis. Additional plots for this section will use a zoomed in axis in order to more easily show the differences in stratification.





After we plotted a general Kaplan-Meier curve, we began producing plots for specific variables, to determine whether we could detect differences in how quickly participants get into shelter, depending on specific variables. We found the following variables showed interesting, and potentially meaningful differentiation:

- Program
- Gender
- Age group
- `Sensitive_f` (A variable with sensitive information, for purposes of this report, the actual variable name is masked)
- Having a prior stay with C@P
- `Sensitive_h` (A variable with sensitive information, for purposes of this report, the actual variable name is masked)

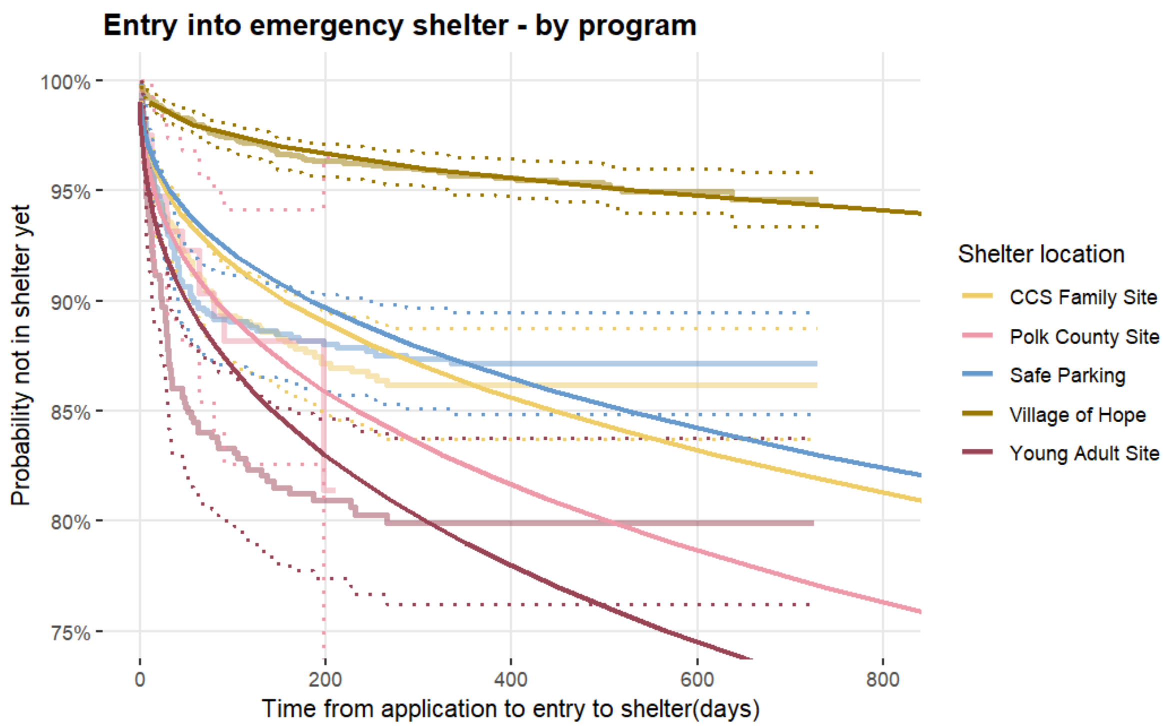
The following plots show a parametric Weibull duration analysis model, plotted over the descriptive Kaplan-Meier curve, and the Kaplan-Meier curve's confidence interval.

9.2.1.1 Shelter Location

In general, it appears that applicants for the Village of Hope (VOH) program tend to have to wait longer to get into shelter, compared to the other four programs. Though our Weibull models may not be a good fit for the other four programs, it appears to fit the VOH data well, and stays within its confidence intervals.

Based on the coefficients of the model, the event time ratio (ETR) indicates how much time it takes for a covariate group to enter shelter, compared to the model's baseline. In this case, the ETR shows that applicants to Village of Hope take about 19 times longer to enter shelter, compared to CCS Family Site. ETRs for other locations are much closer to CCS. For Safe Parking, the ETR is 1.18, which means it takes those applicants about 1.18 times longer to enter shelter. For Young Adult site, the ETR is 0.32, which means it takes those applicants about 32% of the time it takes CCS applicants to enter shelter. These ratios are corroborated by p values for the covariates. P values for the Polk County Site and Safe Parking are 0.32 and 0.63, respectively, indicating there is not a significant difference between those curves and CCS. However, for VOH and Young Adult Site (p values <2e-16 and 0.0012), there is a meaningful difference from CCS (a p value of less than 0.05 indicates a statistically significant difference).

Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
0.41	2.41	CCS Family	Polk County site: -0.64 Safe Parking: 0.16 Village of Hope: 2.96 Young Adult Site: -1.14	Polk County site: 0.32 Safe Parking: 0.63 Village of Hope: <2e-16 Young Adult Site: 0.001

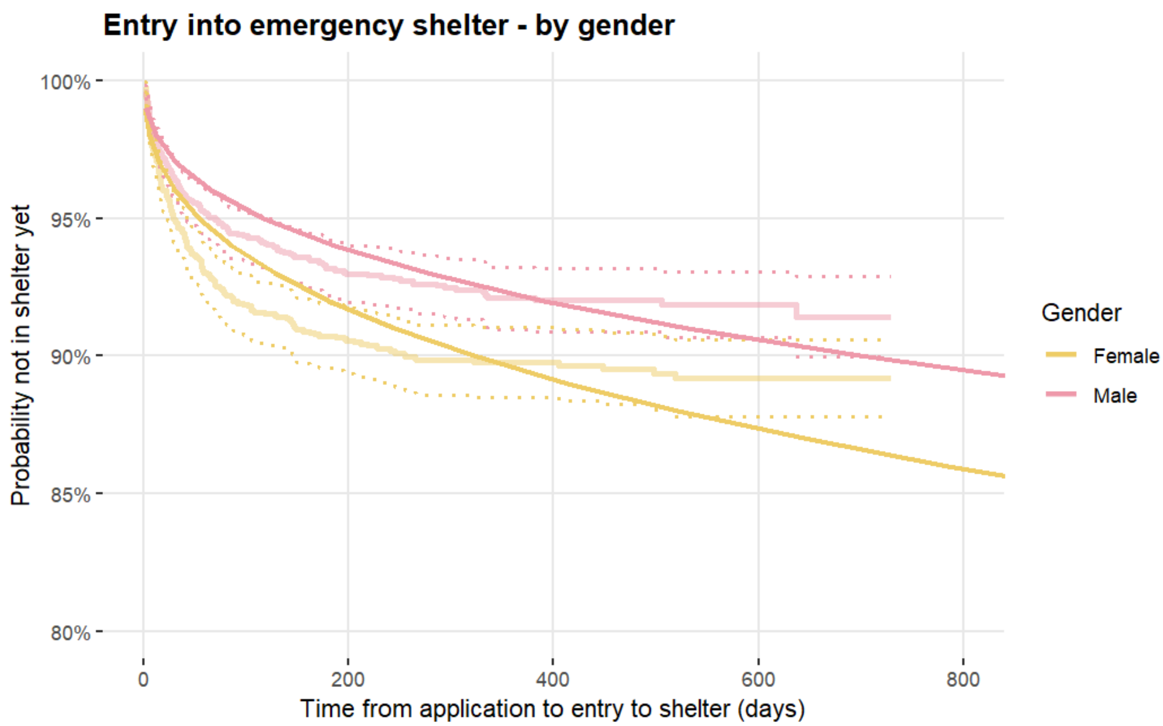


9.2.1.2 Gender Identity

Female applicants appear to enter shelter faster than male applicants. Additionally, the Weibull model appears to have be a decent fit, as it generally stays within the confidence intervals of the Kaplan-Meier plot, with some slight deviations. Though, it appears to fit better for male applicants than female applicants.

Examining the ETR for male applicants, the ETR is 2.17, which indicates male applicants take 2.17 times longer to enter shelter, compared to female applicants. The p value for the male applicants is 0.002, which indicates there is a significant difference between male and female applicants in how quickly they enter shelter.

Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
0.40	2.48	Gender - Female	Male: 0.7728	Male: 0.002

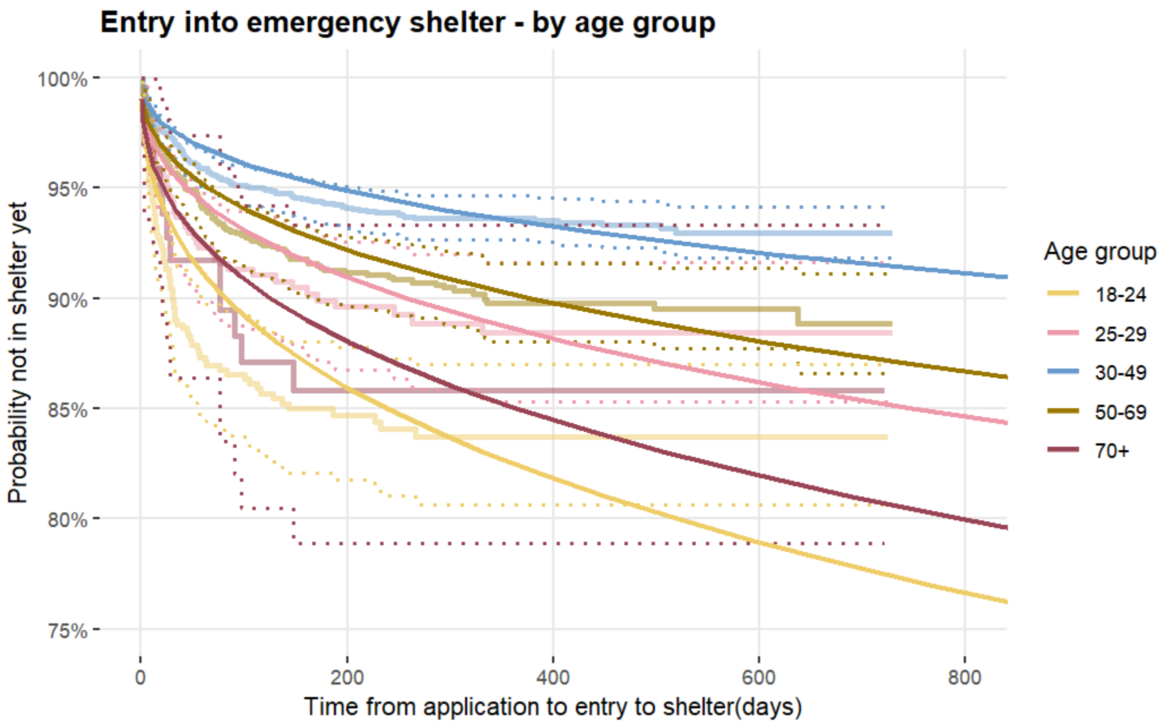


9.2.1.3 Age Groups

Age may come into play into how quickly an applicant enters shelter. We noted significant differences when comparing most age groups to the Weibull model's base line (ages 18-24). However, it should be noted that what we see in this plot may also be a reflection of availability. In conversation with C@P, they indicated there is more availability for the young adult shelter (YAS) compared to others. Additionally, though this model does include applicants age 70+, they comprise a small percentage of all age groups (only 98 people out of 5,108).

Examining the ETRs for older age groups, we see that applicants older than age 24 fall into two groups, in terms of how long they take to enter shelter. For ages 25-29, 50-69, and 70+, ETRs range from 1.53 (70+) to 4.59 (50-69). In contrast, for applicants ages 30-49, their ETR is 13.25, meaning those applicants take about 13 times longer to enter shelter, compared to those who are 18-24. P values for the age groups also reflect statistically significant differences compared to the model baseline for most groups, as shown in the table below. However, we can see there is not a statistically significant difference between applicants ages 18-24 and 70+, since the p value for the 70+ applicants was 0.56.

Weibull Shape	Weibull Scale	Model Baseline	Covariate coefficients	P values
0.41	2.46	Ages 18-24	Ages 25-29: 1.14 Ages 30-49: 2.58 Ages 50-69: 1.53 Ages 70+: 0.42	Ages 25-29: 0.01 Ages 30-49: 6.5e-14 Ages 50-69: 1.1e-05 Ages 70+: 0.56

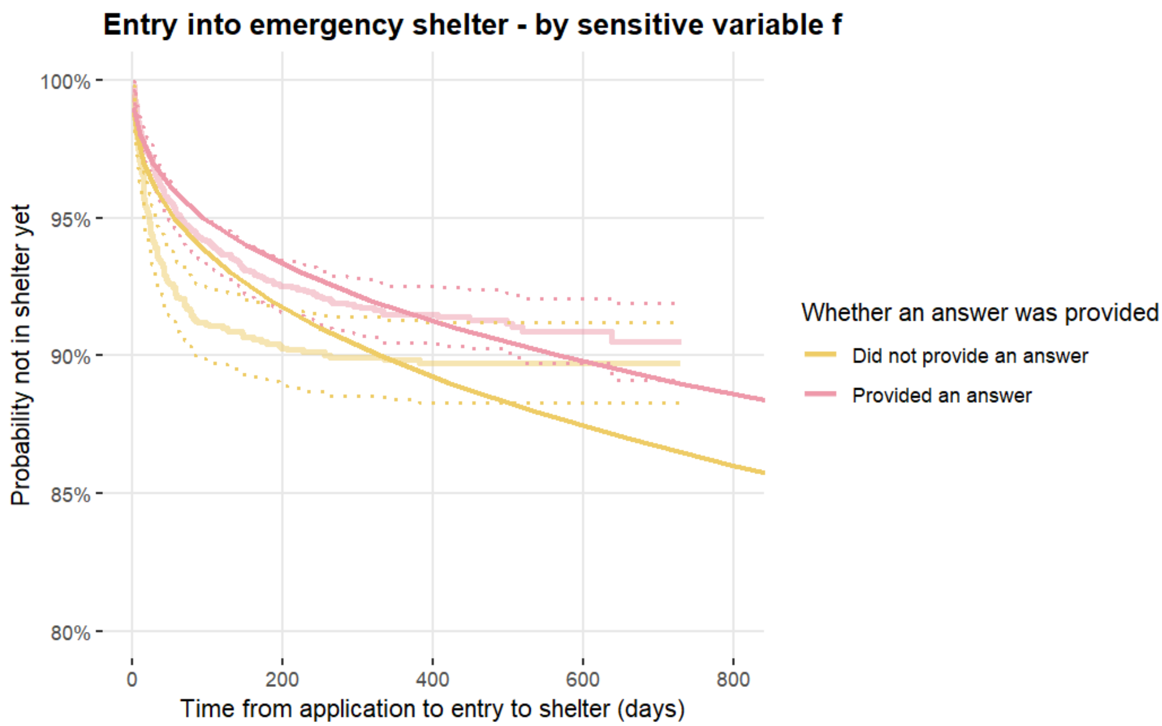


9.2.1.4 `Sensitive_f` Variable

Applicants who provide an answer to this application question tend to take longer to enter shelter. This is consistent with the ETR for that group, which shows they take 1.72 times longer to enter than those who don't provide a response. The p value for those who provided a response also confirms there is a statistically significant difference, as it is 0.03.

Our Weibull model does not fit data as well for those who did not answer, as it does not stay within the Kaplan-Meier confidence intervals very well. In comparison, the model for those who did provide an answer has a better fit.

Weibull Shape	Weibull Scale	Model Baseline	Covariate coefficients	P values
0.40	2.48	Did not provide an answer	Provided an answer: 0.54	Provided an answer: 0.03

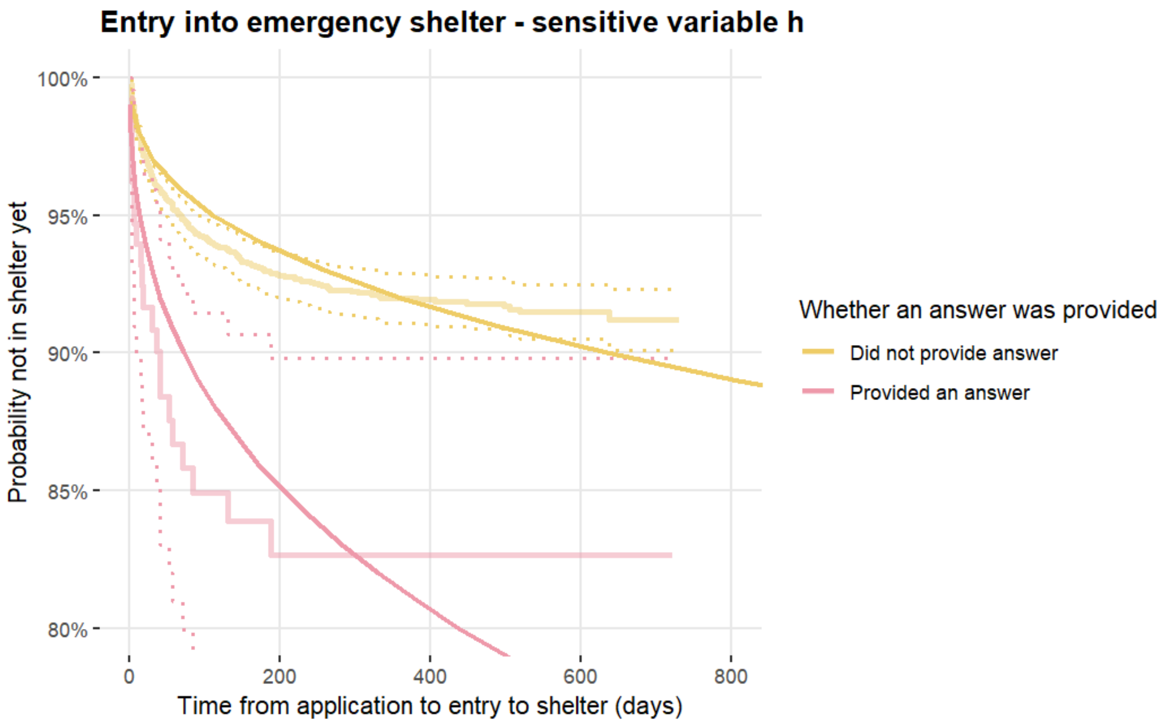


9.2.1.5 `Sensitive_h` Variable

Applicants who provide an answer to this application question tend to enter shelter much faster than those who don't. This is consistent with the ETR for that group, which indicates those who answer only take 11% of the time it takes the no response group to enter shelter. The p value for those who provided a response also confirms there is a statistically significant difference, as it is well below 0.05 (see table below).

In terms of fit, the Weibull model stays well within the confidence intervals for those who provided an answer, though the confidence interval for that group is clearly much wider than that of the non response group. However, it should be noted that the number of applicants who provided an answer is much fewer than those who did not (132 versus 4140). It's possible a different trend would emerge if the split was more even.

Weibull Shape	Weibull Scale	Model Baseline	Covariate coefficients	P values
0.41	2.41	Did not provide an answer	Provided an answer: -2.18	Provided an answer: 8.4e-05



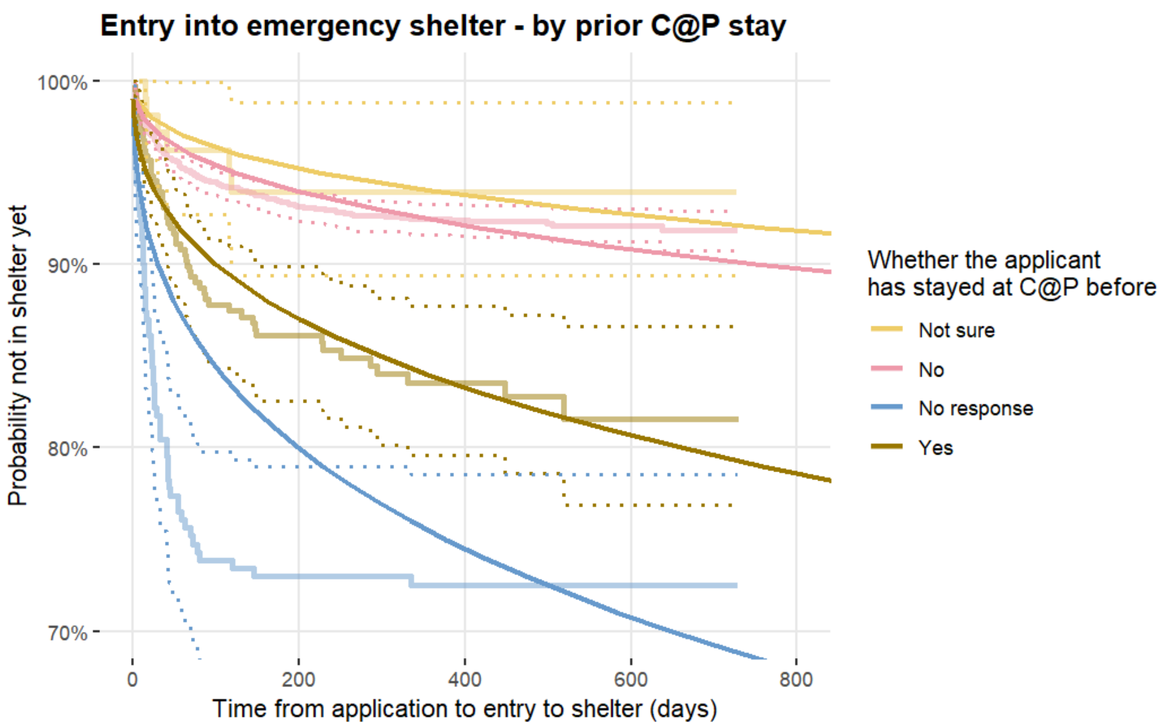
9.2.1.6 Having a prior stay at C@P

We see interesting differentiation in the data when it comes to whether an applicant has stayed with C@P before.

Our Weibull model uses the "not sure" response group as the baseline. Looking at ETRs for the other groups, we see the ETRs for "no", "no response", and "yes", are 0.55, 0.02, and 0.07, respectively. This means that for these groups, at most, they take 55% to 2% of the time that the not sure group takes to enter shelter. P values for these groups tell a similar story. The p values for "no response" and "yes" are 0.0005 and 0.016, respectively, meaning there is a statistically significant difference from the "not sure" group. However, the p value for the "no" group indicates there is no significant difference between "no" and "not sure".

For people who answer "yes", our Weibull model stays well within the Kaplan-Meier confidence intervals and seems to be a good fit. For other answers, the model does not fit the data as well, though even when the model for "no" deviates from Kaplan-Meier confidence intervals, it still remains fairly close. However, it should be noted the distribution of answers is very lopsided. The lion's share (around 4,400) of responses are "no", with "yes" responses (376) being the next largest group.

Weibull Shape	Weibull Scale	Model Baseline	Covariate coefficients	P values
0.40	2.52	Response "not sure"	No: -0.60 No response: -3.82 Yes: -2.62	No: 0.56 No response: 0.0005 Yes: 0.016



9.2.2 HMIS Stays

When we applied duration analysis to data on stays in shelter, we found notable differences for some of the same variables as applications, along with other variables unique to the stays data. We noticed that the variables we found of interest generally fell into two categories: traits or circumstances of the participant or level of engagement with case managers.

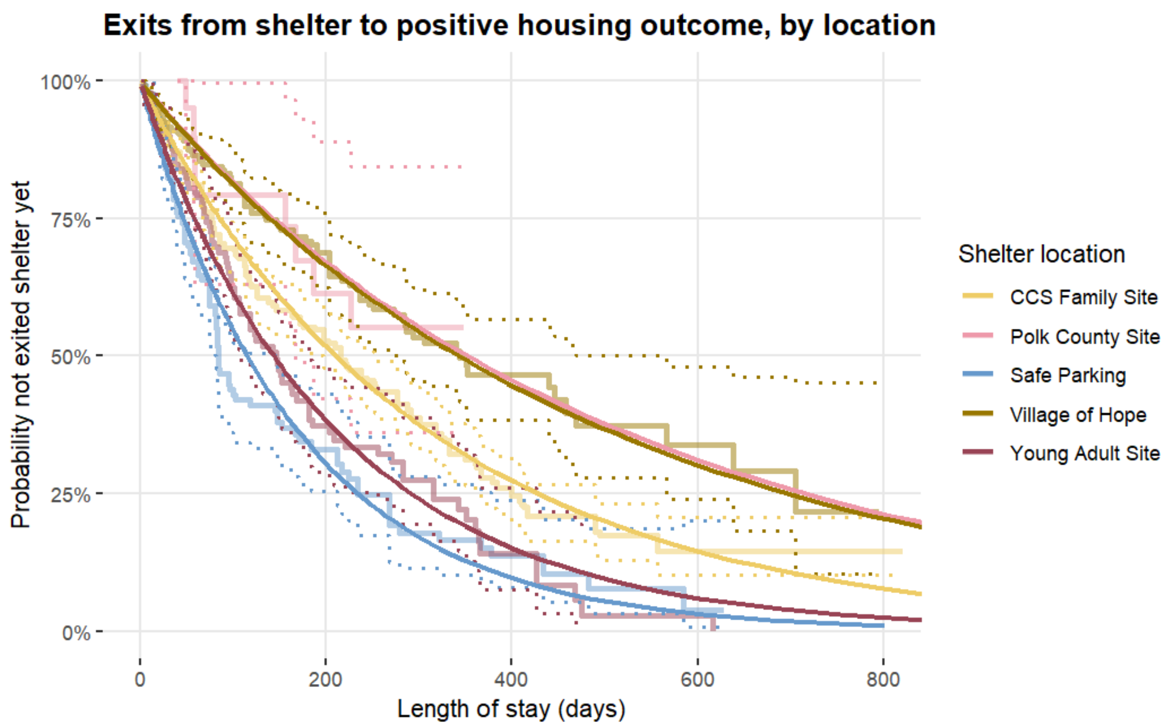
9.2.2.1 Shelter Location

Similar to the application analysis, provider seems to be a factor in how quickly a participant exits to a positive housing outcome. For applications, it generally appeared that it took longer for participants to enter VOH. Similarly, it takes longer to exit VOH to a positive outcome. This difference is reflected mirrored in the plot, and in the p value comparing VOH to the baseline the model used (CCS), which is 0.0001.

Regarding fit, again, our Weibull model appears to be a good fit for VOH, staying within the Kaplan-Meier confidence intervals. In comparison to the applications, the model does appear to fit most of the other program location data as well, with a possible exception for the Polk County site. The Polk County site had only 12 stays, so even though the model fit appears good, that could change as more data is collected over time. For the non VOH locations, the model appears to stay within the Kaplan-Meier confidence intervals, but there is clearly some overlap in those intervals.

ETRs for the programs show that differences in the time to exit shelter are not as stark compared to the time to enter shelter. The greatest difference we see is between Village of Hope, Polk County site, and CCS. Village of Hope and Polk County site participants take 1.63 (VOH) and 1.67 (Polk) times longer to exit shelter compared to CCS. On the other end of the spectrum, Safe Parking participants take the least amount of time to exit shelter, taking 55% less time to exit than CCS participants.

Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
0.98	1.02	CCS Family	Polk County site: 0.51 Safe Parking: -0.60 Village of Hope: 0.49 Young Adult Site: -0.38	Polk County site: 0.16 Safe Parking: 1.7e-07 Village of Hope: 0.0001 Young Adult Site: 0.002

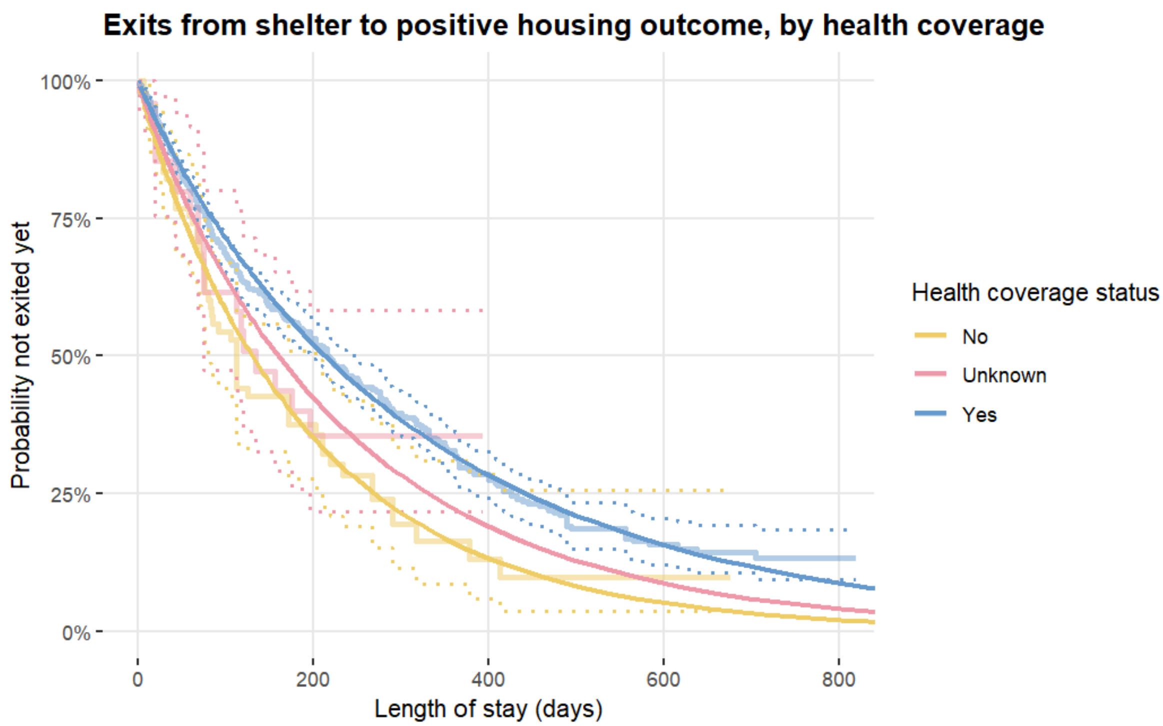


9.2.2.2 Participant Health Coverage

Having health coverage appears to play a role in staying in shelter longer. Our Weibull model indicates that the difference between answering yes and no is statistically significant with a p value below 0.0007. Additionally, for the "yes" and "no" groups, the Weibull model appears to fit well and generally stays within the Kaplan-Meier confidence intervals.

The ETR for the "yes" group indicates that those participants take about 1.64 times longer to exit shelter compared to the "no" group. Similarly, participants in the "unknown" group take about 1.23 times longer.

Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
0.95	1.05	No	Unknown: 0.21 Yes: 0.50	Unknown: 0.44 Yes: 0.0007



9.2.2.3 Gender Identity

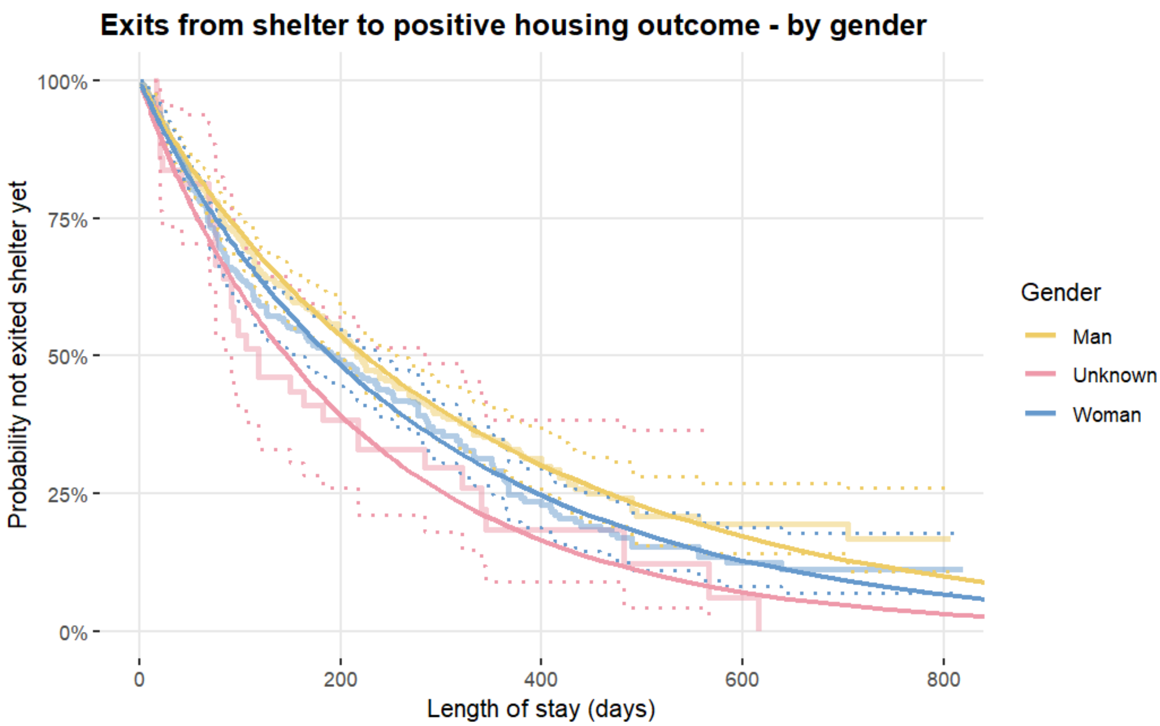
For the applications analysis, we had a statistically significant finding that women tend to enter shelter faster than men. For our analysis of stays, and how quickly participants exit shelter, our findings are slightly different.

Our stays model uses men as the baseline, and we can see that the p value for women is 0.07, which indicates there is no statistically significant difference between men and women in terms of how quickly they enter shelter. Similarly, the ETR for women is 0.85, meaning that women take about 85% of the time men take to leave shelter.

For the unknown group, we do note that there does appear to be a statistically significant difference between the unknown group and men, with a p value of 0.03 for the unknown group. However, it should be noted that there are very few participants that are in this group (only 47 compared to over 500 for both men and women)

In terms of model fit, because all of the groups are clustered much closer together, we also see the confidence intervals from the Kaplan-Meier curves overlapping. However, visually, the curves do appear quite close to the data as modeled by the Kaplan-Meier curves.

Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
0.94	1.06	Gender - Man	Unknown: -0.4314 Woman: -0.1643	Unknown: 0.03 Woman: 0.07

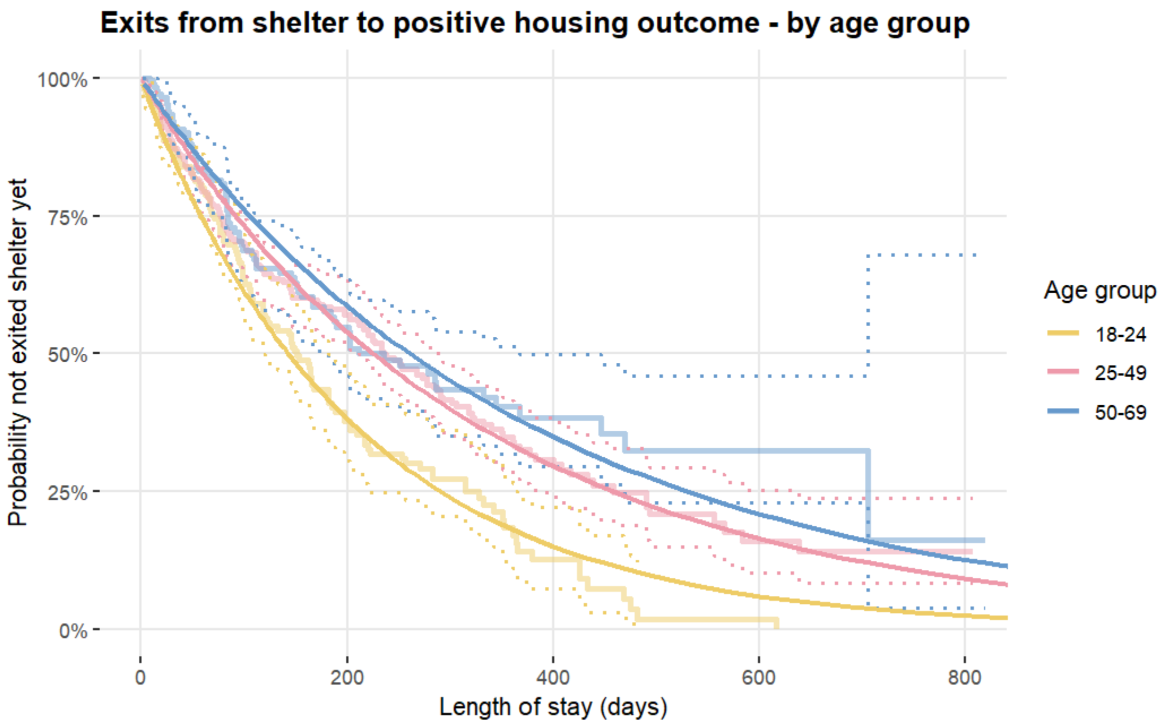


9.2.2.4 Age Groups

Our application analysis found that younger participants (18-24) tend to enter shelter faster than other groups. For our analysis of stays, our Weibull model uses ages 18-24 as the baseline and similarly shows that participants ages 18-24 leave shelter faster than older age groups. Based on ETRs for the other groups, participants 25-49 and 50-69 take 1.58 (25-49) to 1.82 (50-69) times longer to exit shelter. P values for these groups (shown in the table below) corroborate this finding, indicating there is a statistically significant difference between participants older than age 24 compared to those 18-24.

Regarding goodness of fit, the plot below shows that the Weibull model seems to fit the data well and stay well within the Kaplan-Meier confidence intervals. The models for the other two age groups also seem to fit fairly close with the data, but it is also clear visually there is not much difference between participants ages 25-49 and 50-69. The confidence intervals for those groups also overlap, making those groups harder to distinguish from each other.

Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
0.98	1.02	Ages 18-24	Ages 25-49: 0.45 Ages 50-69: 0.61	Ages 25-49: 0.0001 Ages 50-69: 6e-05



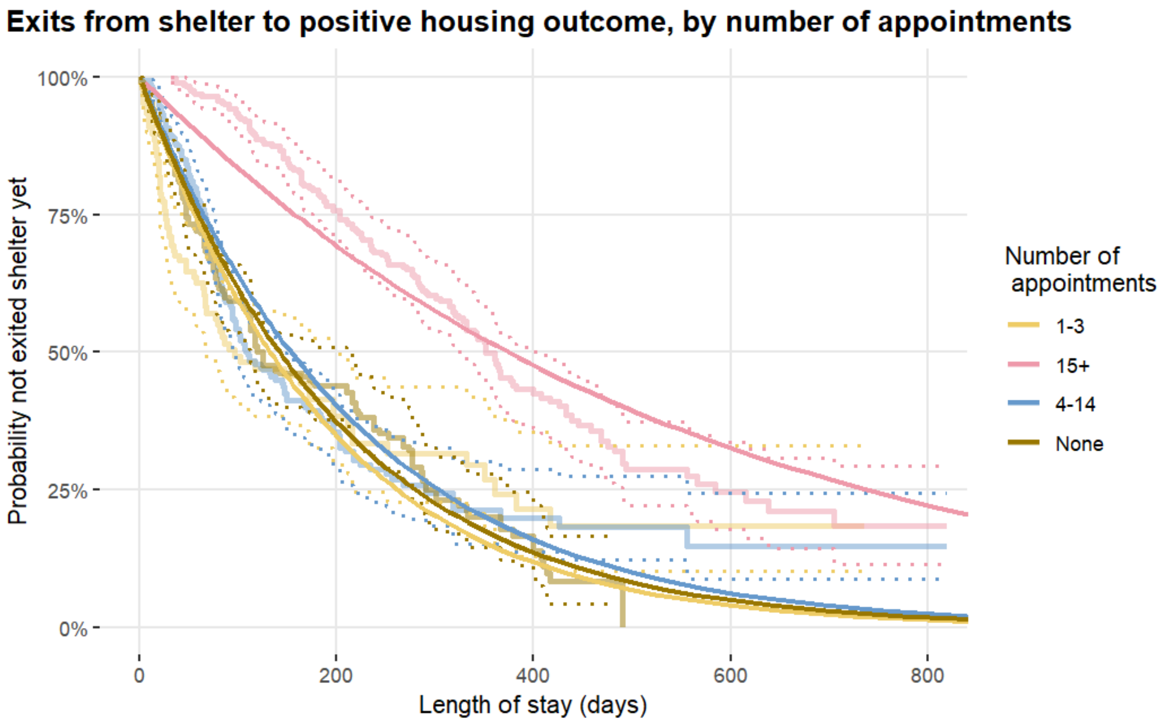
9.2.2.5 Count of Appointments & Appointment Times

We identified an interesting trend in the number of case management appointments a participant had. If the number of appointments was less than 14, participants exited shelter faster compared to participants with 15 or more appointments.

Our Weibull model uses the 1-3 appointment group as the baseline. Visually, the plot shows that the models for 4-14 and zero appointments are very close to the 1-3 appointment group. The p values for the zero and 4-14 appointment groups are consistent with this, as they are above 0.05, meaning there was no statistically significant difference from the 1-3 appointment group. However, for the 15+ appointment group, the p value is well below 0.05, and the ETR 2.82. Thus, the 15+ appointment group is notably different than the 1-3 appointment group and takes 2.82 times longer to exit shelter.

Though none of our Weibull curves perfectly fit the underlying data, it does appear that generally, the curves for the lower number of appointments stay within the confidence intervals for the lower appointment groups, and do not overlap much with the intervals for the 15+ group.

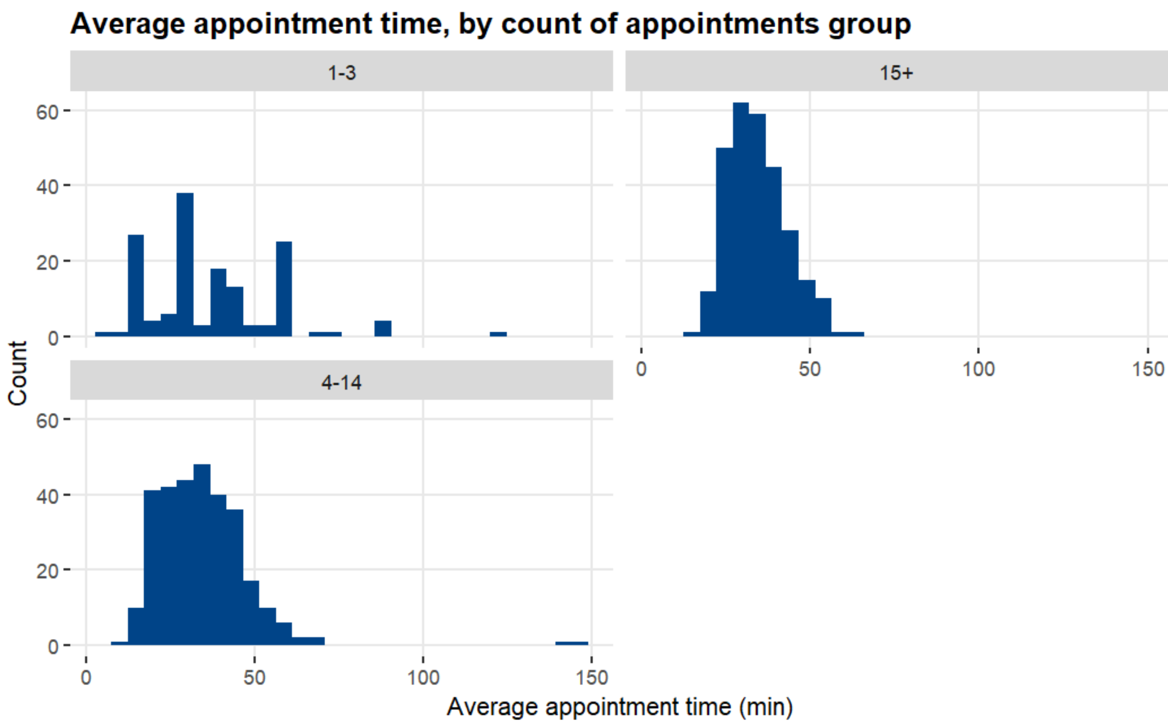
Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
1.02	0.98	1-3 appointments	15+ appt.: 1.04 4-14 appt.: 0.15 Zero appt.: 0.07	15+ appt.: 2.5e-14 4-14 appt.: 0.26 Zero appt.: 0.60



Since there is a gulf between participants with 15+ appointments versus those with less, we did some additional investigation to see if we could detect any association between having appointments and positive outcomes. We did so from a few angles.

First, we examined the duration curve for exiting to a *negative* outcome, where we saw the same pattern detected with positive exits. So, regardless of whether a participant exits to a positive or a negative exit, those with more appointments tend to stay longer.

Second, we looked at how much time participants spend at their appointments. We noted that in the duration analysis curve for average appointment time, participants with higher average appointment times tend to leave shelter faster, and generally, the curves for all times are clustered fairly closely together. However, when we examine the distribution of average appointment time, participants with 4-14 or 15+ appointments have a similar and consistent range of appointment times, while the range for participants with 1-3 appointments was much more inconsistent and varied.

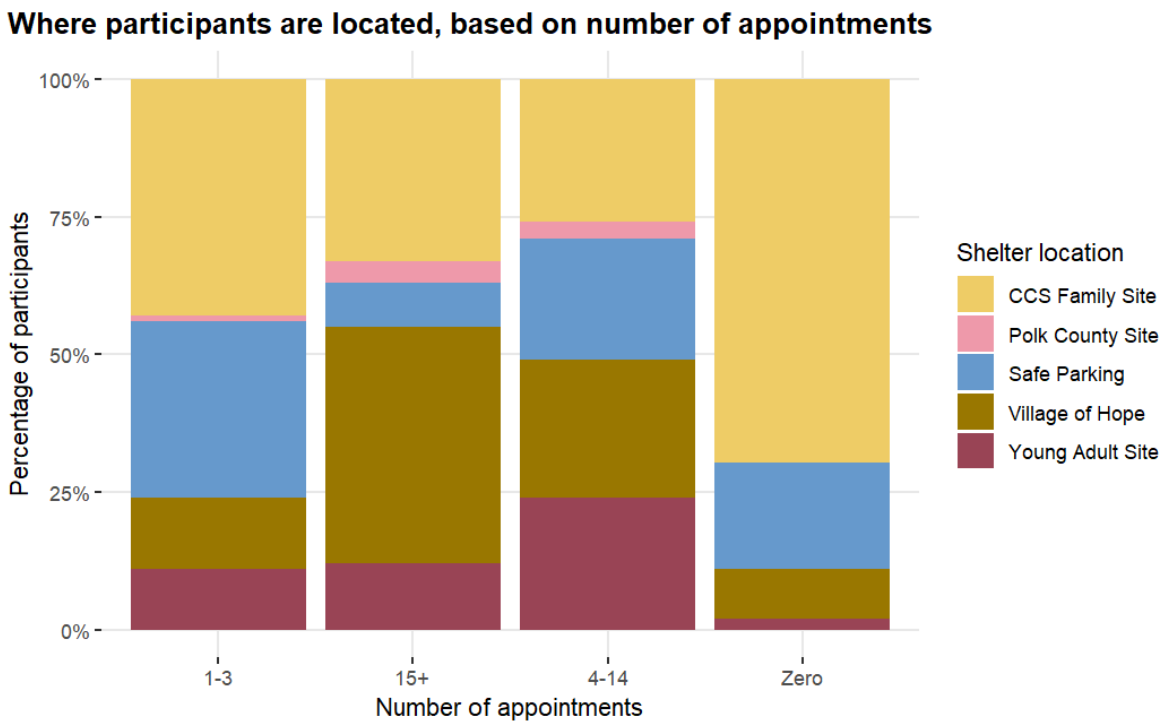


Third, we looked at how many participants exit to a positive outcome, depending on how many appointments they had. We found that 53% percent of participants with 4-14 appointments exited to a positive outcome, and 52% percent of participants with 15+ appointments exited to a positive outcome. In contrast, only 43% of participants with 1-3 appointments exited to a positive outcome.

Focusing on the 1-3 appointment group more, we found the proportions for where those participants came from were notably different from the other appointment groups. As shown below, for each appointment group, the composition of where participants are located varies from group to group. For the 1-3 appointment group, it's notable that it is the group with the highest percentage of participants in the Safe Parking program (32.09%).

For the 15+ appointment group, we also see it is the group with the highest percentage of participants in Village of Hope. This seems consistent with our other finding that participants from VOH tend to take longer to exit shelter.

Lastly, we also noted that the zero appointments group has a significantly higher proportion of participants from CCS. However, that's likely because CCS is a family shelter, so that data includes children that are unlikely to be attending case management appointments. Inclusion of children in CCS data may be inflating the number of participants in the zero appointments group.

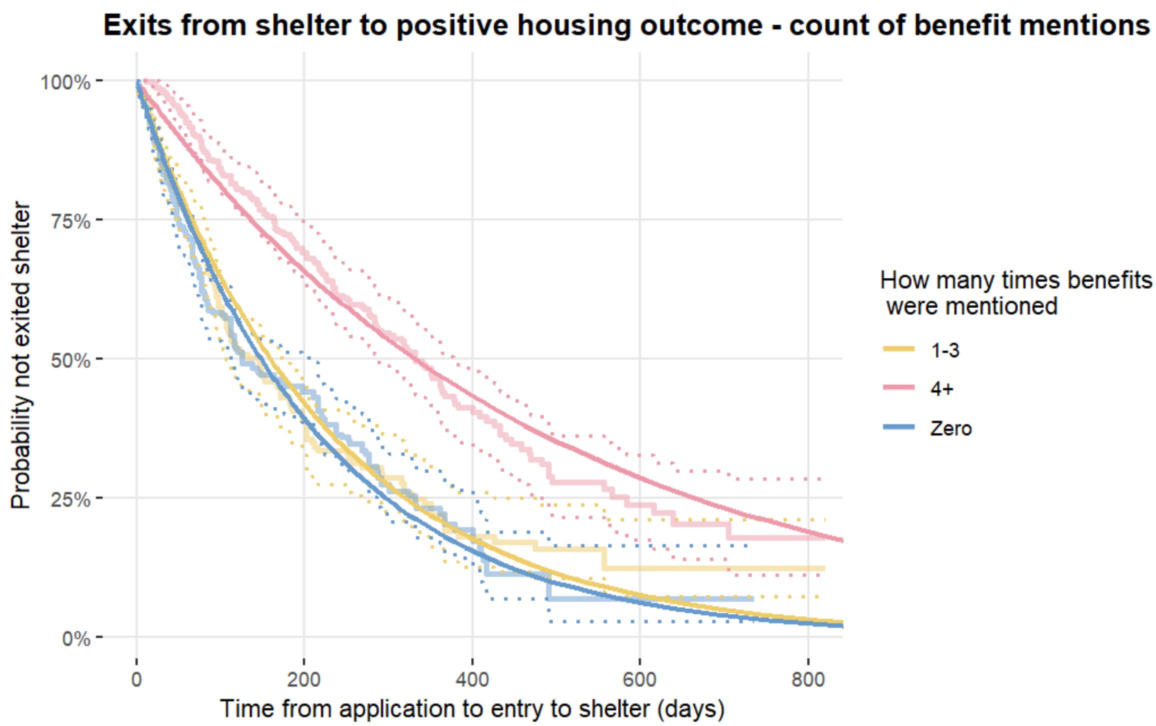


9.2.2.6 Counts of Specific Words & LDA Topics

Our dataset contained variables that we found might be proxies for participant engagement with case management, rather than direct indicators. These proxy variables indicated how many times certain words appeared in case management notes, or counted how many times a topic appeared in case notes. In general, we found that more counts of the specified words (medication, identification, benefits, appointments), and topics seemed tied to longer stays in shelter. Below is an example of the differentiation we observed for one of these variables (specifically, the number of times the word "benefit" was mentioned).

For this model, the 1-3 mentions group was the baseline, and the p values for the 4+ or zero mentions indicate there is a statistically significant difference between the 1-3 mentions and 4+ mentions group. However, there is not a statistically significant difference between 1-3 mentions and zero mentions. For participants with 4+ mentions, their ETR is 2.08, meaning they take about 2 times as long to exit shelter compared to those with 1-3 mentions. Similar to what we saw for the count of appointments model, we see that our model fits the data well and tends to stay within the Kaplan-Meier confidence intervals. However, we can't distinguish much between the participants who have zero mentions versus those with 1-3.

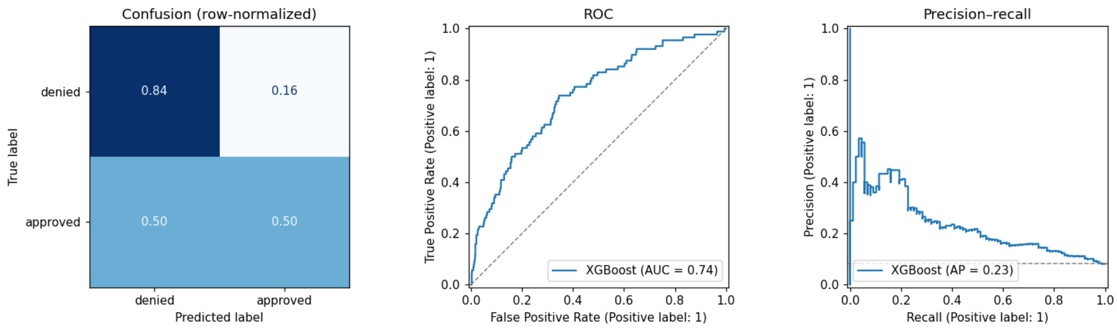
Weibull Shape	Weibull Scale	Model baseline:	Covariate coefficients	P values
1	1.00	1-3 mentions	4+ mentions: 0.73 Zero mentions: -0.07	4+ mentions: 2.4e-11 Zero mentions: 0.46

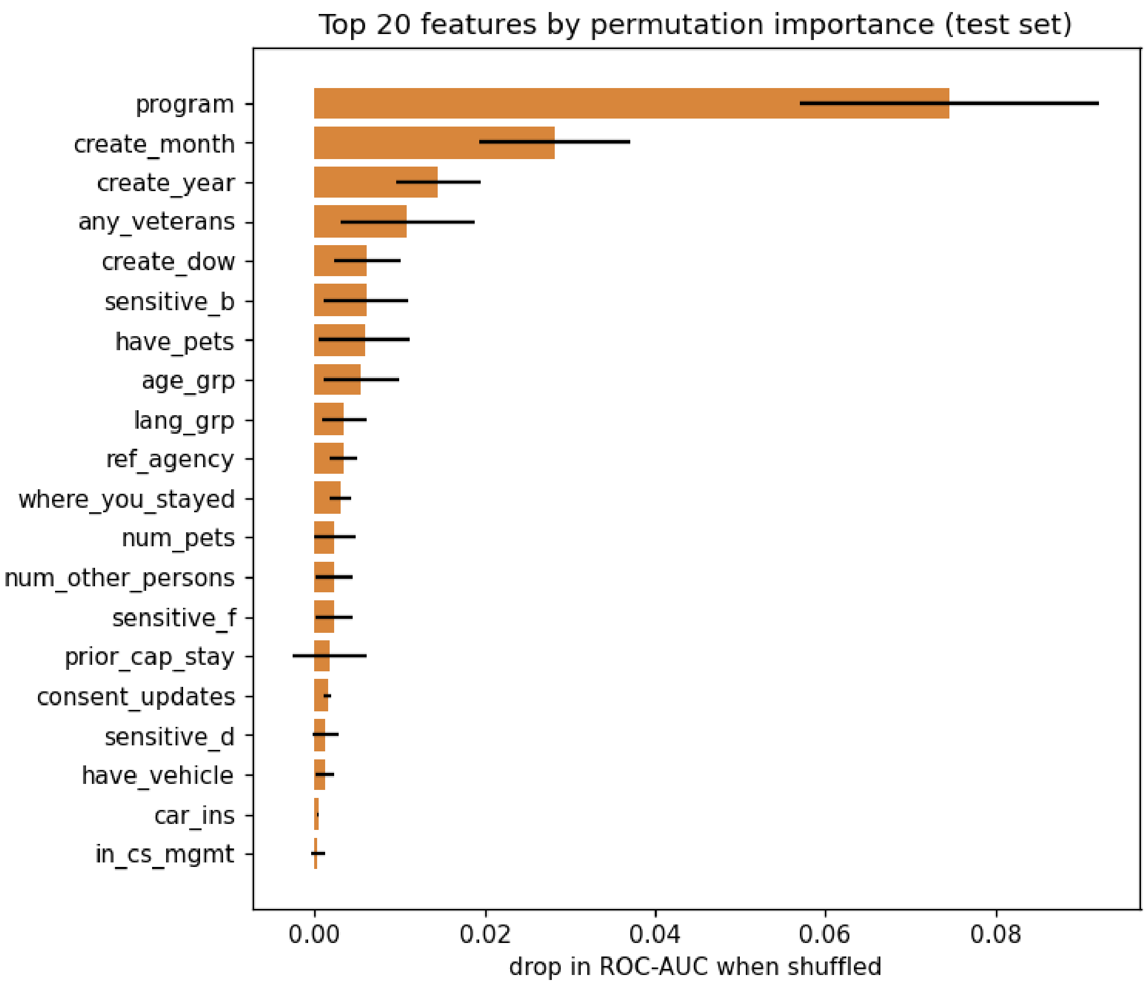


9.3 Machine Learning

9.3.1 Entry to Shelter

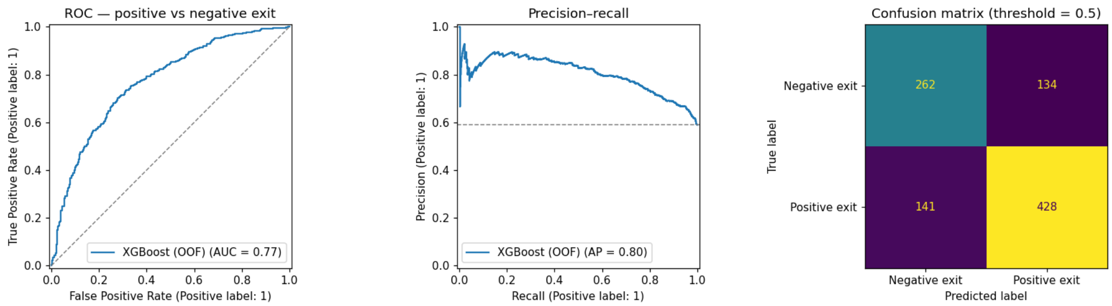
While this model appeared to be somewhat predictive of whether a participant would successfully enter shelter, we found that the top 2 features by permutation importance were the program they applied to, and the month of their application. This suggests that external factors, namely the availability of shelter at a particular site at a point in time, are more predictive than traits of the participant. Since the application numbers are relatively small for some of the programs, we were not able to create models for each program individually.





9.3.2 Exit Type from Shelter

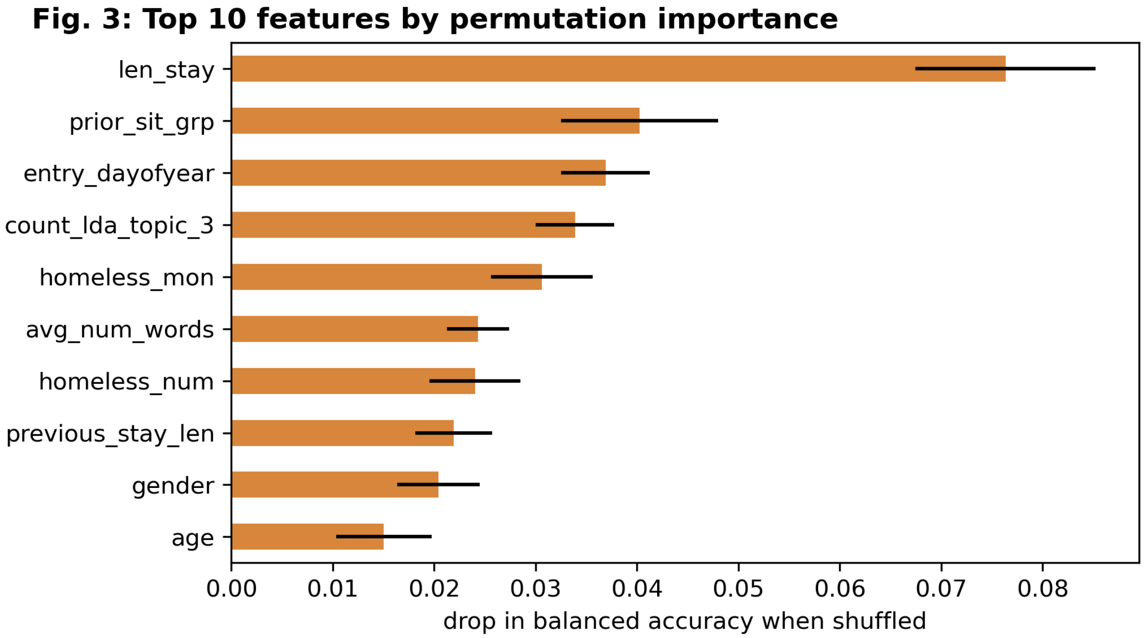
The model was evaluated using the following metrics:



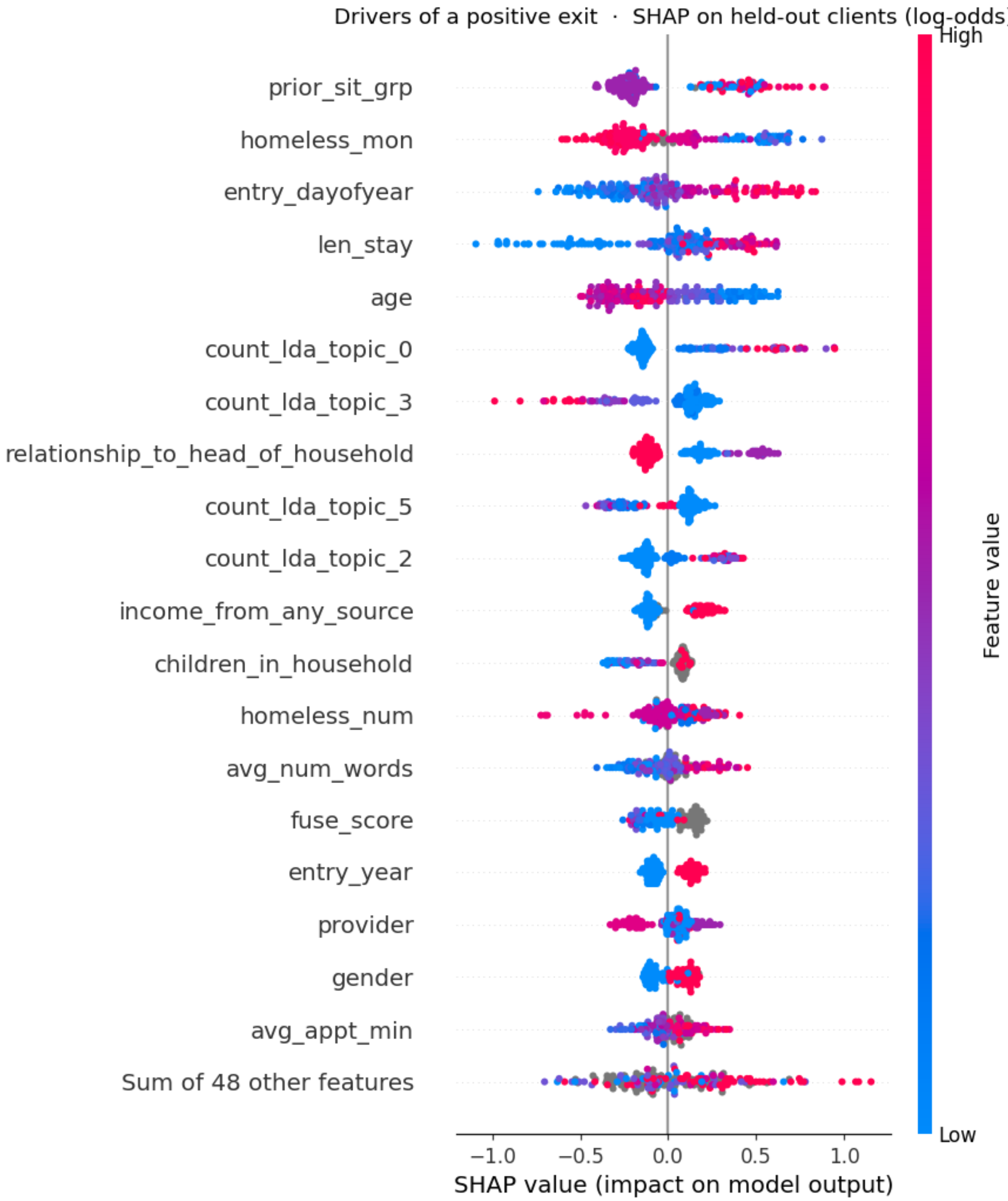
- Pooled out-of-fold AUC: 0.78
- Accuracy: 0.726
- Balanced accuracy: 0.73
- Class-level precision and recall:
  - Positive class: Precision = 0.78, Recall = 0.75
  - Negative class: Precision = 0.66, Recall = 0.70

Fold-level metrics were relatively stable (AUC ranging from 0.75-0.81 across folds), suggesting the result isn't driven by a single fold. The precision and recall metrics suggest that the model is somewhat better at predicting positive exits than negative ones, consistent with the class imbalance.

This next graph demonstrates the effect on removing various variables from the XGBoost classifier model and its overall change in accuracy when not included. Length of Stay is the most prominent variable in this model with an accuracy change of 8%.



Finally, we evaluate individual variable effects by examining the following SHAP graph.



9.3.2.1 Length of Stay

This feature has a wide spread of importance to the model, from strongly negative to mildly positive. Short stays split between neutral and negative, while long stays behave more uniformly for a positive exit.

9.3.2.2 Months Homeless

This feature tends to be predictive of a negative exit outcome. Participants who have been homeless for a longer period of time tend to have a higher likelihood of exiting to a negative outcome, compared to those who have been homeless for a shorter period of time. Note that there *are* positive values for this feature for some cases.

9.3.2.3 Age

This feature has a fuzzy split at around 30 years old. Participants younger than 30 tend to have a higher likelihood of exiting to a positive outcome, compared to those who are older than 30.

9.3.2.4 Prior Living Situation

This feature has mixed importance to the model, with some prior situations being more predictive of a positive exit, and others being more predictive of a negative exit.

9.3.2.5 LDA Topic 3

This feature was generated from the case management data, and is a topic that seems to be related to management and safety of the participant's pod - likely an indication of participants who are struggling to adhere to rules or guidelines. This feature tends to be negatively predictive of a positive exit outcome.

10 Discussion

Our results provide insight into factors that seem to influence specific metrics (length of time to wait for shelter, stay in shelter, returning to shelter, or successfully obtaining assistance). As discussed in the results section, we found a mix of factors can affect these metrics. Combining our two (duration analysis and machine learning), we believe these are the primary high level take ways:

- There is a significant gulf between the number of people who apply for shelter, and the number of people who ultimately enter shelter. We believe this is likely linked to a limited supply and availability of shelter, and demonstrates a significant need for shelter in the Salem area community.

- Both how quickly a person enters shelter, and whether they exit shelter to a positive outcome appears to have some association with the shelter location, the participant's gender, and their age.
- Longer stays are associated with a positive exit from shelter, and higher level of case management engagement is associated with longer stays in shelter.

However, during our analysis and partnership with C@P, we also identified real world circumstances and limitations of the data that affect interpretation of these results.

10.1 CCS Missingness and Outcomes

When completing duration analysis, we noted that for many variables, participants who provided no response exited shelter faster than those who provided a definitive response (e.g., "yes" or "no"). We observed for many of those cases, CCS made up a higher proportion of the non-responses, compared to other answers. In discussion with C@P, we learned that this might be due to the nature of the shelter and C@P's data collection. We learned that the assessments the shelter stay data are based on are a substantial effort to fill out, and since CCS is a family shelter, completing these assessments for whole families is a much bigger effort compared to other C@P programs. We also learned that assessments are conducted for children, but that the questions are not tailored for children. So, there are a number of questions that are simply not relevant or answerable for a child (for example, a question about connecting to VA services). Together, these two factors may be contributing to the high amount of non-responses from CCS and affecting the results for time to exit shelter.

10.2 Temporality & External Factors

10.2.1 Applications

In our analysis of applications, though our machine learning model did not have much predictive power, it did surface some factors that may affect how application data is analyzed and interpreted. Based on the results from that model, we discussed with C@P how season can have an impact on how quickly applications are processed, and how availability of shelter comes into play as well. Regarding the latter item, of note for our study was the Polk County site, which opened in June 2025. When that site opened, for the initial participants, it was likely faster to enter shelter, compared to those who applied later once the shelter had been in operation longer.

In general, this makes it difficult to answer what factors are associated with successful entry into shelter. And more specifically, determine whether entry is due traits of the participant versus external circumstances (e.g., availability of beds, timing of application). Our difficulty in creating a model to predict successful entry may be a result of the gap between community need and the available assistance.

10.2.2 Stays

Many factors can influence the length of time a participant stays in shelter. One in particular is the availability of local housing units. C@P has noted a number of recent examples of new developments opening, which allow many C@P residents to move into housing, opening up space for new participants to enter C@P programs. On the flip side, if other programs, such as housing vouchers, become less available, that can also have an impact on how quickly participants are able to leave shelter. These factors currently are not present in our data and would be difficult to incorporate into future analysis. However, they are important to consider when interpreting the results of our analysis.

10.3 C@P's Philosophy & Significance of Results

One aspect of this project looked at how quickly certain events happen (entry into shelter, or exiting shelter). For the latter, though we have made observations about what seems to increase or decrease the length of stay in shelter, we do not believe one should conclude that an increase is always bad, or that a decrease is always good. From C@P's perspective, they would rather have participants stay in shelter, than leave to a negative outcome. Additionally, C@P noted that in some cases, a participant may be ready to move out of shelter but are simply waiting on assistance (such as a housing voucher) to do so. So, when considering the length of stay in shelter, it's important to consider that it may be influenced by outside factors, not necessarily the individual's readiness. Additionally, altering the length of stay is not necessarily the desired outcome.

10.4 Limitations/Future Analysis

Our discussion touches on some caveats on this analysis, but some additional ones we identified as follows.

10.4.1 Scope & Causation

We did not examine what factors lead to specific outcomes. For example, this analysis doesn't cover what factors specifically lead to obtaining rental housing versus living with a friend. Our analysis is not causative, and more work would be needed to establish what factors (if any) cause specific outcomes or actions.

10.4.2 Generalization

This study only uses data for one organization, in one specific part of the state. Our findings may not be highly generalizable to other populations.

10.4.3 Data Quality

We found that connecting a participant across data sources was difficult, and we likely did not match every participant across all three sources. To maintain participant confidentiality, we matched applicants to outcomes and case management data within SharePoint, which limited our matching process to what could be done in the version of Excel available in SharePoint. Additionally, we had limited data points to match an individual on (name, date of birth, email, phone number).

During the matching process, we found that not all participants could be matched to an application. This could have been due to data quality issues (for example, omissions, misspellings), and due to the fact that we could not easily implement a fuzzy matching process within Excel. Procedural and data entry improvements to identify a participant across all stages of C@P's processes could improve the quantity and quality of the data collected.

10.4.4 Manual Processes

Some of the data collection and matching processes were manual, which could introduce errors or inconsistencies. Automating these processes where possible could improve data quality and reliability. The mandated systems for data collection (HMIS and Activate Care) do not allow for easy export of data, and require manual processes to extract data. This is a limitation of the data collection systems, and not a limitation of C@P's data collection practices.

10.4.5 Independence of Observations

The data from C@P does not include any flag that reliably indicates whether a participant is a part of a family unit that is in shelter together, or which family unit they belong to. This means that in our shelter stay dataset, the outcome of one individual can be linked to or influences another. For example, both the parents and children of a family can show up in CCS data, and the outcome for the parent is likely going to be mirrored for the children. This means that our duration analysis does not strictly meet an assumption of independence between individuals, it least for CCS data.

11 Conclusion of Recommendations & Future Work

11.1 General Recommendations for Future Work

- Conduct similar studies for other organizations, to see if other organizations identify similar, or different factors that affect entry to and exit from shelter.

- Test entry models for the programs individually, in the cases where they have sufficient data for modeling. This may help identify factors that are more specific to a particular program, and may be masked when looking at all programs together.
- Conduct more complex duration analysis, to investigate whether differences can be detected in combinations of factors, rather than individual factors. This could build a better understanding of the populations C@P serves and how entry or exit time varies depending on the participant's circumstances.
- Additional machine learning analysis to cluster or classify applications data, to assist with triaging applications for housing or waitlist opportunities. This may help C@P with prioritizing applications when there are many applicants that *could* be moved into emergency shelter, if there were ones available. This may help C@P pursue funding opportunities, if the analysis shows that there is high need, beyond just the volume of applications. (For example, being able to demonstrate that C@P assisted a certain number of participants, but that there were other participants with similar situations and applications C@P was unable to serve due to limited capacity).
- With more time and more data available, future analyses could be more granular and program specific. For example, there are several types of outcomes that fall under a positive outcome, and doing analysis on more specific outcomes, or a particular program, could yield more tailored insights.

11.2 Recommendations for C@P

- We found that compared to other programs, it takes longer to enter into the Village of Hope shelter. To improve access to shelter for this population, we would recommend investing resources to expand that site, or create another, similar one. To our understanding, that is something C@P is already embarking on.
- Investing more resources and funds in Safe Parking may be beneficial for a few reasons:
  - C@P has noted that participants in Safe Parking often move into other C@P programs. Since Safe Parking is likely one of the programs with the lower barriers to entry, and lower cost to administer, expanding this program is one of the less resource intensive ways to improve access to services. Additionally, since Safe Parking may provide some increased stability for participants, it may put them in a better position to have a positive outcome if they ultimately move from Safe Parking to another C@P shelter.
  - Participants who engaged in case management more (4+ appointments) exit to positive outcomes more often compared to those with 1-3 appointments. Additionally, we noticed that participants with 1-3 appointments tend to have inconsistent average appointment times, and that a higher proportion of these participants come from the Safe Parking program. For participants at Safe Parking, with only 1-3 appointments, increasing case management resources, availability, or follow-up might improve their chances of exiting to a positive outcome.

- As noted in the discussion section, we observed that we likely did not match every participant to an application. If C@P modified data management practices to create a consistent identifier for a participant across all three data sources, it would make future analyses easier and potentially more robust. For example, establishing a process where all HMIS IDs are ultimately added to the application data would help immensely in following a participant's journey with C@P from beginning to end. Similarly, finding a way to capture within C@P data whether a person belongs to a family unit, and which family unit they belong to, opens up opportunities to do analysis not just of individuals but of families.
- During clean up of the application data, we noted some modifications to the application programming would help keep the application data tidier and more readily usable for analysis. For example:
  - Altering the date of birth field so the current date cannot be entered, and so the applicant cannot be below a certain age (e.g., the applicant can't be 1 year old).
  - Providing standardized answers, instead of free text. For example, for language group, providing a drop down of standardized options would simplify the data for that field. A free text field could be provided for participants who wish to select a language not listed.
- We also noted many of the applications have similar or overlapping questions. Consolidating the separate applications into one form, with programmed logic to indicate specific programs of interest, could make the data from applications easier to organize and analyze later on.
- We recommend that C@P continue to invest in data collection and analysis, in particular retaining historical program data, and ensuring that data is collected in a consistent manner across programs. This will allow for more robust analysis in the future, and allow C@P to monitor trends over time. Clear documentation of changes to data collection practices (such as updating forms or case note fields), and changes to programs, will also help with future analysis and interpretation of results.
- For future analysis and reporting, we recommend that C@P invest time in iterating a data pipeline that can be run on a regular basis, to produce updated reports and visualizations. This would allow C@P to monitor trends over time, and identify changes in the population they serve, or in the outcomes of their programs. A first version of this pipeline has been created, and is available in our GitHub repository. Subsequent iterations could include Excel macros to automate the manual processes of exporting data from HMIS and Activate Care, identity resolution between datasets, and cleaning the data for analysis. Additionally, future iterations could include a periodic combined dataset (eventually a data warehouse), to store the data from all three sources in one location, and allow for easier access and analysis.

11.3 Overall Conclusions

Our integrated data helped identify factors that may influence a participant's experience, but they do not convey the full story of what affects the metrics we examine. Our findings should be considered as a stepping stone for understanding how participants move through C@P's programs, and should be considered side by side with staff and participant experiences. With more robust data gathering practices and deeper examination of key variables, we believe that greater insights can be found to consistently and effectively assist C@P's data analysis efforts.

12 Reproducibility and References

12.1 Reproduce Main Results

Our scripts for analysis are contained in our GitHub repository, along with data dictionary and general data pipeline information. Detailed information on manual processes has been saved in C@P's SharePoint, and is not shared publicly, since it contains information about sensitive data fields that have been masked to ensure participant privacy. If this project were to be reproduced, interested parties would need to work with C@P to obtain the data, access data in SharePoint, and follow our notes on the manual cleaning. Once the data is ready for export, our Python and R scripts can be ran to process the data and produce analyses.

12.2 Dependencies

Libraries we used were as follows:

- Python: pandas, numpy, seaborn, scikit-learn, duckdb, matplotlib, pathlib, xgboost, shap, re
- R: tidyverse, janitor, lubridate, survival, survminer, rlang, broom, stringr, ggally

12.3 Data

Data can only be obtained with permission from C@P. Interested parties must contact C@P to inquire about data access and future projects.

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